Merits of Matplotlib Transcript

Welcome to BITS PLANET DIGITAL. Today we will continue our learning journey for the course Data Visualisation Storytelling. We will be now moving to module number 4 wherein you will be using various software tools for building your data visualisations.

In this regard, we will be first looking at the Python software wherein you will be trying to understand the various features in Python and use them for making your visualisations. In Python, as you are being exposed to the Python environment in your course, you would be using various libraries for arriving at certain functions. We will be looking at the specific libraries and modules within Python framework that help you build effective visualisations.

In this regard, we will be first looking at Mathplotlib, then Seaborn and advanced features like Bokeh. These libraries help you to customise your visualisations and deliver the information to your audience. In this lesson, it will be an experiential or live demo kind of a thing wherein we will just have a quick intro to Mathplotlib and then move on to the interface how to use Mathplotlib.

We will be now exploring the basic features of Mathplotlib. I will give an overview. Then we will move on to the live demo session.

So, just a brief original about Mathplotlib. Mathplotlib was created by a neurobiologist called by name John Hunter who was working with the EEG data, the electrocardiogram data. Then he tried to develop this into a software that can be used in different fields.

Today, Mathplotlib is a widely used open source library within the Python ecosystem wherein you can leverage its features for building visualisations. It has its origins in the MATLAB environment, but it is now more or more accustomed to the object orientated programming within the Python interface. So, the way that we will be learning this particular module as opposed to the other three modules, we will be doing more on hands-on while trying to teach the concepts because this is not a theory kind of an environment where everything can be structured.

Having said that, we will provide you with the basic contours of how to explore all these softwares, be it Python, Power BI or Tableau. But it is up to the student to the depth to which they want to learn these software tools and the features therein. So, we will be giving you the resources and explaining the basic features or the features that we think are more relevant, but that is only a guidance.

These are vast tools. These tools have vast applications and very high potential. So, I would advise the students in this module to practise more and try to explore the different features.

So, we will point you towards those resources where you can get an understanding of the features, but it is up to you to really test their performance on different data sets and bring out different visualisations. So, I'm just taking you to the Mathplotlib environment. So, if you click on the link that I just gave you or just type Mathplotlib in any browser, you would go to this particular page.

There is a quick start guide. There is a user guide, tutorials, examples. We'll be often referring to this particular section.

So, familiarise yourself with this particular environment of Mathplotlib. This is the direct source so that you can directly use the codes and the examples given there for your practising. So, now let us see the quick start guide.

So, Mathplotlib, as you know, is a plotting software within the Python ecosystem. It's a library. So, you can import that as Mathplotlib.pyplot. So, we will see this and it is used for presenting graphics.

This is a simple line plot and drawn using the Mathplotlib features. Then, of course, then you have what are the different types of access, features that you can use for bringing pre-attentive attributes or enhancing the visual appeal. So, then you have colours, how to use text, so on and so forth.

We will be seeing some of these features. But as I said, the extent of the software or the features of this tool is very, very wide. So, you can really spend some time on each of these features as you want and then get across the content and use them in your examples or for designing visuals.

So, there are tutorials which you can pick up and then start learning. Then there is examples, what are the different types of charts that you can make use of your Mathplotlib.

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